

# IBM University Engagement Project Submission

*“Exploring the Power of Agentic AI with IBM Granite and IBM BOB”*

# Project Tile – Agentic Career Counseling Companion

**Domain of Project** – Education Technology , Agentic AI

| Student Name | Recent Passport Photo                                                              | Email ID                 | Mobile number [WhatsApp] |
|--------------|------------------------------------------------------------------------------------|--------------------------|--------------------------|
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# Problem Statement

## **Fitness Buddy.**

### **The Challenge**

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to lack of personalized guidance, time constraints, and inconsistent motivation. Traditional fitness solutions often require expensive subscriptions, in-person consultations, or rigid schedules that don't adapt to personal preferences or daily routines. There is a growing need for an accessible, friendly, and intelligent virtual assistant that can provide on demand fitness advice, healthy lifestyle suggestions, and basic nutrition guidance—all tailored to individual needs and available at any time. Fitness Buddy aims to solve this problem by offering a conversational, AI-powered health and fitness .

# Proposed Solution -

**Fitness Buddy** is an AI-powered virtual health and fitness assistant designed to provide users with personalized fitness guidance anytime and anywhere. Built using **IBM Granite** on **IBM Cloud Lite**, the application interacts with users through a conversational interface to understand their fitness goals, lifestyle, and preferences. Based on the user's inputs—such as age, height, weight, fitness objective, available workout time, dietary preferences, and activity level—the AI generates customized home workout plans, healthy meal suggestions, hydration recommendations, and daily motivational messages.

The system also helps users build long-term healthy habits by setting daily fitness goals, tracking workout progress, monitoring BMI, and maintaining consistency through reminders and achievement streaks. Unlike traditional fitness applications that often require expensive subscriptions or fixed schedules, Fitness Buddy delivers personalized, on-demand guidance that adapts to each user's needs. By leveraging the natural language capabilities of IBM Granite, users can ask fitness-related questions in simple conversational language and receive accurate, easy-to-understand recommendations.

The solution is designed to be accessible, affordable, and user-friendly, making professional fitness guidance available to anyone with an internet connection. Through continuous interaction and personalized recommendations, Fitness Buddy encourages users to adopt healthier lifestyles, stay motivated, and achieve their fitness goals effectively.

# Technology Used

## 1. IBM Cloud Lite

IBM Cloud Lite serves as the cloud platform for developing, hosting, and deploying the Fitness Buddy application. It provides a free and scalable environment to integrate IBM AI services, manage application resources, and ensure secure access to the chatbot and backend services.

## 2. IBM watsonx.ai Studio

IBM watsonx.ai Studio is used to build and integrate AI capabilities into the application. It enables access to IBM Granite foundation models, allowing Fitness Buddy to understand user queries and generate personalized workout plans, nutrition recommendations, motivational messages, and healthy lifestyle suggestions through natural language interactions.

## 3. IBM Granite (Foundation Model)

IBM Granite is the core large language model that powers the intelligent conversational assistant. It analyzes user inputs such as fitness goals, age, activity level, and dietary preferences to provide context-aware, personalized responses. Granite delivers customized home workout routines, meal suggestions, habit-building advice, and daily motivation, creating a smart and interactive fitness coaching experience.

#### **4. IBM Business Object Builder (BOB)**

IBM BOB is used to simplify the development and orchestration of the application by connecting AI services, business logic, and user interactions. It helps streamline workflows, manage application components, and improve the overall efficiency of deploying and maintaining the Fitness Buddy solution.

#### **5. Frontend Technologies**

HTML5

CSS3

JavaScript

Bootstrap

These technologies are used to create a responsive, user-friendly interface where users can interact with the Fitness Buddy chatbot, view workout plans, track fitness progress, and receive personalized recommendations.

#### **6. Backend Technology**

Python (Flask)

Flask is used to handle user requests, process application logic, communicate with IBM watsonx.ai services, and manage data flow between the frontend and IBM Cloud services.

# Project Output - Dashboard

 **Fitness Buddy**

QUICK TOPICS

-  Home Workout
-  Meal Plan
-  Daily Motivation
-  HIIT Training
-  Strength Training
-  Pre/Post Workout
-  Build Habits
-  Hydration Tips
-  Yoga Routine

YOUR STATS

0

MESSAGES

1

SESSIONS

Powered by **IBM Granite**  
IBM Watsonx.ai

 **Fitness Buddy** AI Coach

Clear



**Hey there! I'm Fitness Buddy**  
Your personal AI-powered health & fitness coach, available 24/7. Ask me anything about workouts, nutrition, or motivation!

 Home Workouts

 Meal Ideas

 Motivation

 Fitness Plan

Ask about workouts, nutrition, motivation...



Powered by IBM Granite · IBM Watsonx.ai · Not a medical service

# Daily Fitness and Diet

 **Fitness Buddy**

QUICK TOPICS

 Home Workout

 Meal Plan

 Daily Motivation

 HIIT Training

 Strength Training

 Pre/Post Workout

 Build Habits

 Hydration Tips

 Yoga Routine

YOUR STATS

4

MESSAGES

1

SESSIONS

Powered by **IBM Granite**  
IBM Watsonx.ai

 **Fitness Buddy** AI Coach

YOU 

Suggest a healthy meal plan for weight loss for the week.

02:37 pm

 **FITNESS BUDDY**

Here are some quick, nutritious meal ideas:

**Breakfast** 

- Overnight oats with banana & chia seeds
- Greek yogurt with berries & honey

**Lunch** 

- Grilled chicken wrap with avocado & greens
- Lentil soup with whole-grain bread

**Dinner** 

- Baked salmon with steamed broccoli & quinoa
- Stir-fried tofu with vegetables & brown rice

Ask about workouts, nutrition, motivation...



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QUICK TOPICS

-  Home Workout
-  Meal Plan
-  Daily Motivation
-  HIIT Training
-  Strength Training
-  Pre/Post Workout
-  Build Habits
-  Hydration Tips
-  Yoga Routine

YOUR STATS

2

MESSAGES

1

SESSIONS

Powered by IBM Granite  
IBM Watsonx.ai

YOU 

Give me a full-body 20-minute home workout routine.

02:37 pm



FITNESS BUDDY

Here's a quick 20-minute home workout you can try right now!

Warm-up (3 min)

- Jumping jacks — 1 min
- Arm circles + hip rotations — 1 min
- High knees — 1 min

Main Circuit (x3 rounds)

- 15 Push-ups
- 20 Bodyweight squats
- 30-second Plank
- 15 Glute bridges

Ask about workouts, nutrition, motivation...



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# Chat Advisor

 **Fitness Buddy**

QUICK TOPICS

 Home Workout

 Meal Plan

 Daily Motivation

 HIIT Training

 Strength Training

 Pre/Post Workout

 Build Habits

 Hydration Tips

 Yoga Routine

YOUR STATS

14

MESSAGES

1

SESSIONS

Powered by **IBM Granite**

IBM Watsonx.ai

 **Fitness Buddy** AI Coach

YOU 

diet plan for 70 kg person

02:58 pm

 **FITNESS BUDDY**

Here are some quick, nutritious meal ideas:

**Breakfast** 🥗

- Overnight oats with banana & chia seeds
- Greek yogurt with berries & honey

**Lunch** 🥗

- Grilled chicken wrap with avocado & greens
- Lentil soup with whole-grain bread

**Dinner** 🍲

- Baked salmon with steamed broccoli & quinoa
- Stir-fried tofu with vegetables & brown rice

Ask about workouts, nutrition, motivation...



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# IBM BOB

File Edit Selection View Go Run Terminal Help

fitnessbuddy

Upgrade your plan

EXPLORER

FITNESSBUDDY

backend

controllers

routes

services

server.js

frontend

public

src

node\_modules

.env.example

package-lock.json

package.json

README.md

.env.example

```
1 # IBM Cloud Credentials
2 IBM_API_KEY=15rp9v59Bt0r33XXSDHQP3UDBtY9YAidGHL_Q2Nx-dcf
3 IBM_PROJECT_ID=9ddf1e55-4674-4b28-a3b8-fad197ad6668
4
5
6 # IBM Watsonx.ai endpoint (update region if needed)
7 IBM_WATSONX_URL=https://us-south.ml.cloud.ibm.com
8
9 # IBM IAM Token URL
10 IBM_IAM_URL=https://iam.cloud.ibm.com/identity/token
11
12 # IBM Granite Model ID
13 IBM_MODEL_ID=ibm/granite-13b-chat-v2
14
15 # Server Port
16 PORT=3000
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS BOB FINDINGS

node

PS C:\Users\RAPOLU\VARUN\Downloads\fitnessbuddy> npm start
POST /api/chat 200 68.841 ms - 446
GET / 304 1.553 ms - -
GET /style.css 304 3.291 ms - -
GET /app.js 304 5.137 ms - -
GET /favicon.svg 304 1.032 ms - -
[WARN] IBM\_API\_KEY not configured - serving demo response
POST /api/chat 200 2.506 ms - 446
[WARN] IBM\_API\_KEY not configured - serving demo response
POST /api/chat 200 2.169 ms - 530
[WARN] IBM\_API\_KEY not configured - serving demo response
POST /api/chat 200 1.798 ms - 446

IBM BOB

Pasted text #1 - 14 38.0k / 270.0k 0.914

Todo List 10/10

Scaffold project structure (frontend + ba...

Create backend server with IBM Granite ...

Create fitness-specific prompt engineeri...

Build frontend chat UI with fitness theme

Add workout recommendation logic

Add nutrition guidance logic

Add motivational tips logic

Wire up full-stack communication

Create package.json and dependencies

Final review and validation

IBM Granite Integration

| Component     | Detail                                              |
|---------------|-----------------------------------------------------|
| Model         | ibm/granite-13b-chat-v2                             |
| Auth          | IBM IAM token (auto-cached, auto-refreshed)         |
| API           | Watsonx.ai /ml/v1/text/generation                   |
| Prompt format | < system > / < user > / < assistant > chat template |

13 files changed Undo all Show all

Follow up or start new task Ctrl+N

+ Agent Permissions

Ln 4, Col 1 Spaces: 4 UTF-8 LF Dotenv Bob Settings

The image shows a VS Code editor interface for a project named 'fitnessbuddy'. The Explorer sidebar on the left displays the file structure, including folders for 'backend', 'frontend', and 'public'. The 'app.js' file in the 'public' folder is selected and open in the editor. The code in 'app.js' is a JavaScript file that initializes DOM references for various UI elements like chat window, user input, send button, clear button, typing indicator, welcome card, menu button, sidebar, sidebar close button, and stat messages. It also initializes state variables like history, message count, and loading status. The bottom panel shows the 'TERMINAL' tab with the output of the 'npm start' command, displaying API endpoints and response times. The right sidebar shows the 'Task' panel with a list of tasks and the 'IBM Granite Integration' section.

```
1 /**
2  * Fitness Buddy – Frontend Application
3  * Handles chat UI, API calls, markdown rendering
4  */
5
6 (function () {
7   'use strict';
8
9   // — DOM References —
10  const chatWindow = document.getElementById('chatWindow');
11  const userInput = document.getElementById('userInput');
12  const sendBtn = document.getElementById('sendBtn');
13  const clearBtn = document.getElementById('clearBtn');
14  const typingInd = document.getElementById('typingIndicator');
15  const welcomeCard = document.getElementById('welcomeCard');
16  const menuBtn = document.getElementById('menuBtn');
17  const sidebar = document.getElementById('sidebar');
18  const sidebarClose = document.getElementById('sidebarClose');
19  const statMessages = document.getElementById('statMessages');
20
21  // — State —
22  const history = []; // [{role, content}]
23  let messageCount = 0;
24  let isLoading = false;
25
26  // — Sidebar Toggle —
```

PS C:\Users\RAPOLU VARUN\Downloads\fitnessbuddy> npm start  
POST /api/chat 200 68.841 ms - 446  
GET / 304 1.553 ms - -  
GET /style.css 304 3.291 ms - -  
GET /app.js 304 5.137 ms - -  
GET /favicon.svg 304 1.032 ms - -  
[WARN] IBM\_API\_KEY not configured – serving demo response  
POST /api/chat 200 2.506 ms - 446  
[WARN] IBM\_API\_KEY not configured – serving demo response  
POST /api/chat 200 2.169 ms - 530  
[WARN] IBM\_API\_KEY not configured – serving demo response  
POST /api/chat 200 1.798 ms - 446

Task  
Pasted text #1 - 14 lines  
Context Length 38.0k / 270.0k (14%)  
Task Id 9bce5373876f2b9fec8ec59f2f  
Workspace fitnessbuddy  
Tokens 1 342.4k 1 23.1k  
Cache 1 41.6k 1 300.7k  
API Cost 0.914  
Todo List 10/10  
Scaffold project structure (frontend + ba...  
Create backend server with IBM Granite ...  
Create fitness-specific prompt engineeri...  
Build frontend chat UI with fitness theme  
Add workout recommendation logic  
Add nutrition guidance logic  
Add motivational tips logic  
Wire up full-stack communication  
Create package.json and dependencies  
Final review and validation  
IBM Granite Integration  
13 files changed Undo all Show all  
Follow up or start new task Ctrl+I  
Agent Permissions

# IBM Watsonx.ai studio-Resource usage

IBM watsonx.ai Studio

Search in your workspaces

Upgrade

Varun Rapolu's Account

Sydney

VR

Projects / fitness buddy

Overview

Assets

Jobs

Manage

Start working

Recommended

Add users as collaborators

→

Add data to work with

→

Work with data and models in Python or R notebooks

→

Build machine learning models automatically

→

View all

Collapse

Assets

By all

Assets that you create with tools show here. See all assets, including data assets, on the Assets page.



View all

Resource usage

For this month in this project

0.86<sub>CUH</sub>

Your documentation

Get started with your documentation

You can create and manage documents about work that you do in this project.

Open Documentation editor →

Project history

You created project fitness buddy



# Role of Agentic AI in the solution

Agentic AI plays a central role in **Fitness Buddy** by acting as an intelligent virtual fitness coach that can understand user needs, make informed decisions, and provide personalized guidance without constant human intervention. Unlike a traditional chatbot that only responds to predefined questions, Agentic AI analyzes the user's profile, fitness goals, workout history, and preferences to generate customized recommendations and adapt them over time.

In Fitness Buddy, Agentic AI performs the following functions:

**Personalized Fitness Planning:** Creates customized home workout routines based on the user's age, fitness level, goals, available time, and equipment.

**Intelligent Nutrition Guidance:** Recommends healthy meal plans and dietary suggestions according to the user's preferences and fitness objectives.

**Adaptive Recommendations:** Continuously updates workout intensity and meal suggestions based on user progress, feedback, and consistency.



- **Motivation and Habit Building:** Sends motivational messages, daily reminders, and wellness tips to encourage users to maintain healthy habits and achieve long-term fitness goals.
- **Natural Language Interaction:** Uses IBM Granite to understand user queries in conversational language and provide accurate, context-aware responses.
- **Progress Monitoring:** Tracks user activities such as workouts completed, BMI, water intake, and fitness milestones, then provides insights and suggestions for improvement.
- By leveraging Agentic AI with IBM watsonx.ai Studio and IBM Granite, Fitness Buddy delivers a proactive, personalized, and engaging fitness experience, helping users stay motivated and make healthier lifestyle choices.

# Novelty and Uniqueness

Fitness Buddy stands out from conventional fitness applications by combining **Agentic AI** with **IBM Granite** to deliver a truly personalized and adaptive fitness coaching experience. Instead of providing generic workout plans or static diet charts, the system understands each user's goals, lifestyle, preferences, and progress to generate intelligent recommendations in real time. The AI continuously learns from user interactions and adjusts workout intensity, nutrition advice, and daily goals, making the guidance more effective and engaging.

The solution also emphasizes accessibility by offering professional fitness support without expensive subscriptions or in-person consultations. Users can interact with the AI through natural language, making it easy for people of all ages and fitness levels to receive customized guidance anytime and anywhere.

**Key innovative features include:**

**Agentic AI-based personalized coaching** that proactively plans workouts, nutrition, and daily fitness goals.

**Dynamic workout and meal recommendations** that adapt according to user progress, available time, and fitness objectives.



- **Conversational AI powered by IBM Granite**, enabling natural, human-like interactions instead of rule-based responses.
- **Smart habit-building and motivation**, including reminders, daily challenges, achievement streaks, and personalized motivational messages.
- **Progress tracking and intelligent insights** using BMI, workout history, hydration, and consistency to recommend continuous improvements.
- **Cloud-based deployment on IBM Cloud Lite**, allowing secure, scalable, and accessible fitness assistance from any device.
- By integrating **IBM Cloud Lite**, **IBM watsonx.ai Studio**, **IBM Granite**, and **Agentic AI**, Fitness Buddy delivers an intelligent, proactive, and highly personalized virtual fitness coach that goes beyond traditional fitness applications, making healthy living more achievable, affordable, and engaging.

# Git Hub Link

Github URL - <https://github.com/varun014d/fitnessbuddy.git>

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# Future Scope

Fitness Buddy has significant potential for future enhancements to provide a more comprehensive and intelligent fitness experience. By leveraging advancements in Artificial Intelligence, wearable technology, and cloud computing, the platform can evolve into a complete digital health companion.

## **Future enhancements include:**

**Wearable Device Integration:** Connect with smartwatches and fitness bands to automatically track steps, heart rate, calories burned, sleep patterns, and daily activity.

**AI-Based Exercise Form Analysis:** Use computer vision to analyze users' exercise posture through a smartphone camera and provide real-time feedback to reduce the risk of injuries.

**Advanced Nutrition Planning:** Generate personalized weekly meal plans, calorie tracking, grocery lists, and nutrition recommendations based on health conditions and fitness goals.

**Voice-Enabled Fitness Coach:** Support voice interactions in multiple languages, allowing users to receive workout guidance and fitness advice hands-free.

**Mental Wellness Support:** Include guided meditation, stress management techniques, breathing exercises, and mindfulness sessions to promote overall well-being.

**Predictive Health Insights:** Utilize AI to analyze user progress and provide early recommendations for improving fitness, preventing burnout, and maintaining healthy habits.

- **Community and Social Features:** Enable fitness challenges, leaderboards, group workouts, and progress sharing to increase motivation and user engagement.
- **Integration with Healthcare Services:** Allow users to securely share fitness reports with healthcare professionals or receive personalized recommendations based on medical data.
- **Enhanced Agentic AI Capabilities:** Enable the AI to proactively monitor user progress, adjust fitness plans automatically, set personalized goals, and provide continuous coaching based on long-term performance.
- **Scalable Cloud Deployment:** Expand the platform using IBM Cloud services to support a larger user base, enhanced analytics, and seamless integration with future AI technologies.
- With these enhancements, Fitness Buddy can evolve from a virtual fitness assistant into a comprehensive AI-powered personal health and wellness ecosystem that supports users in achieving long-term fitness and healthy lifestyle goals.

# Certificates Earned: Getting Started with Artificial Intelligence



# Certificates Earned: Troubleshoot Your Code Using IBM Bob

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Varun Rapolu

for the completion of

**Lab: Troubleshoot Your Code Using IBM Bob**

(ALM-COURSE\_4071307)

According to the Adobe Learning Manager system of record

**Completion date:** 04 Jul 2026 (GMT)

**Learning hours:** 30 mins

# Thank You!

Thank you for your time and interest.